MOMENT: A family of open time-series foundation models

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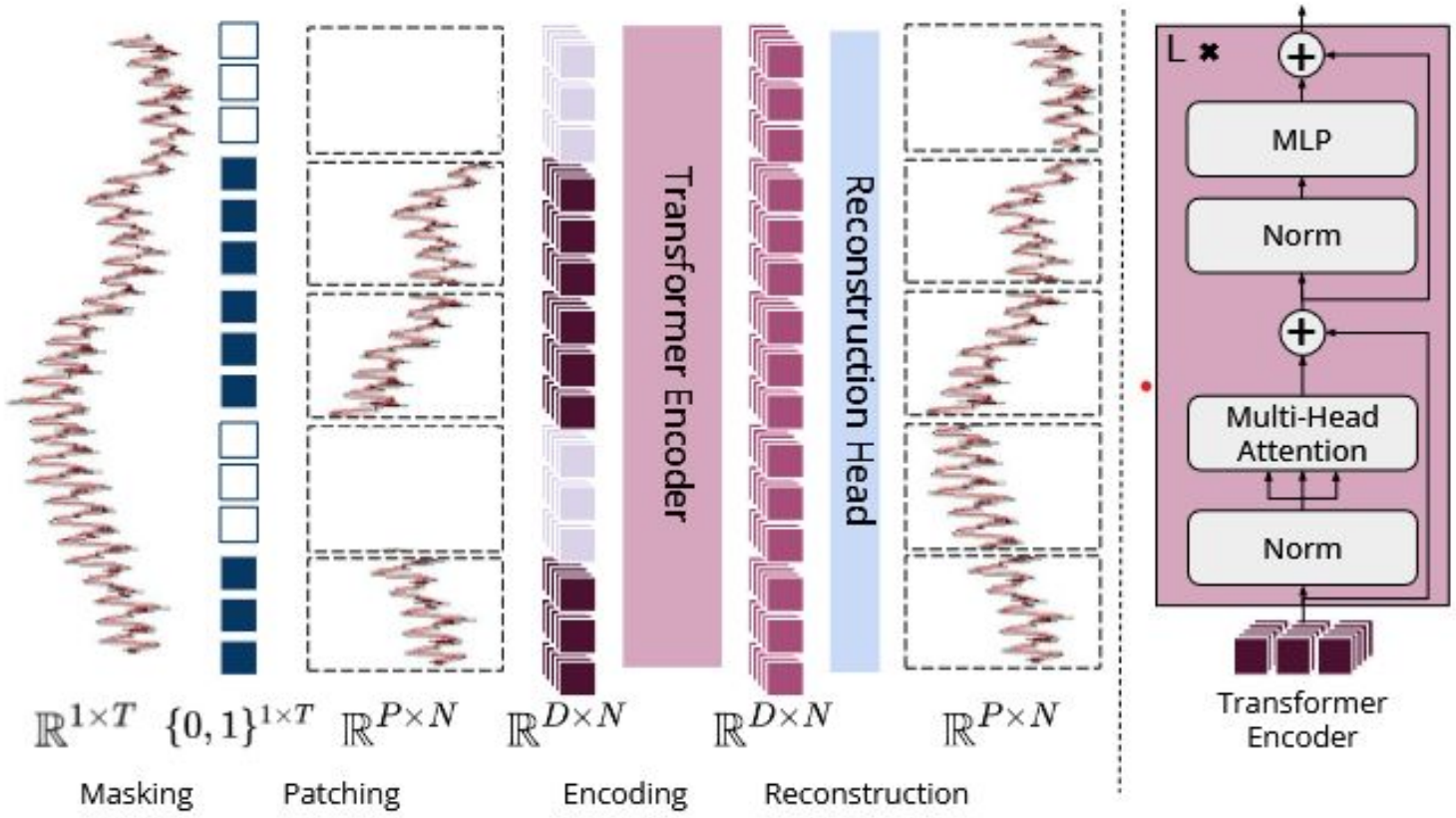
Main contribution

show that pre-training a model of sufficient capacity on a large corpus of unlabeled time series data can in fact enable it to provide reasonably accurate predictions in limited-supervision settings.

Tasks:

1. Long term forecasting
2. Short term foreasting
3. Classification
4. Anamoly detection
5. Imputation

Overview



Masked signal modeling

Transformer based encoder

Lightweight decoder

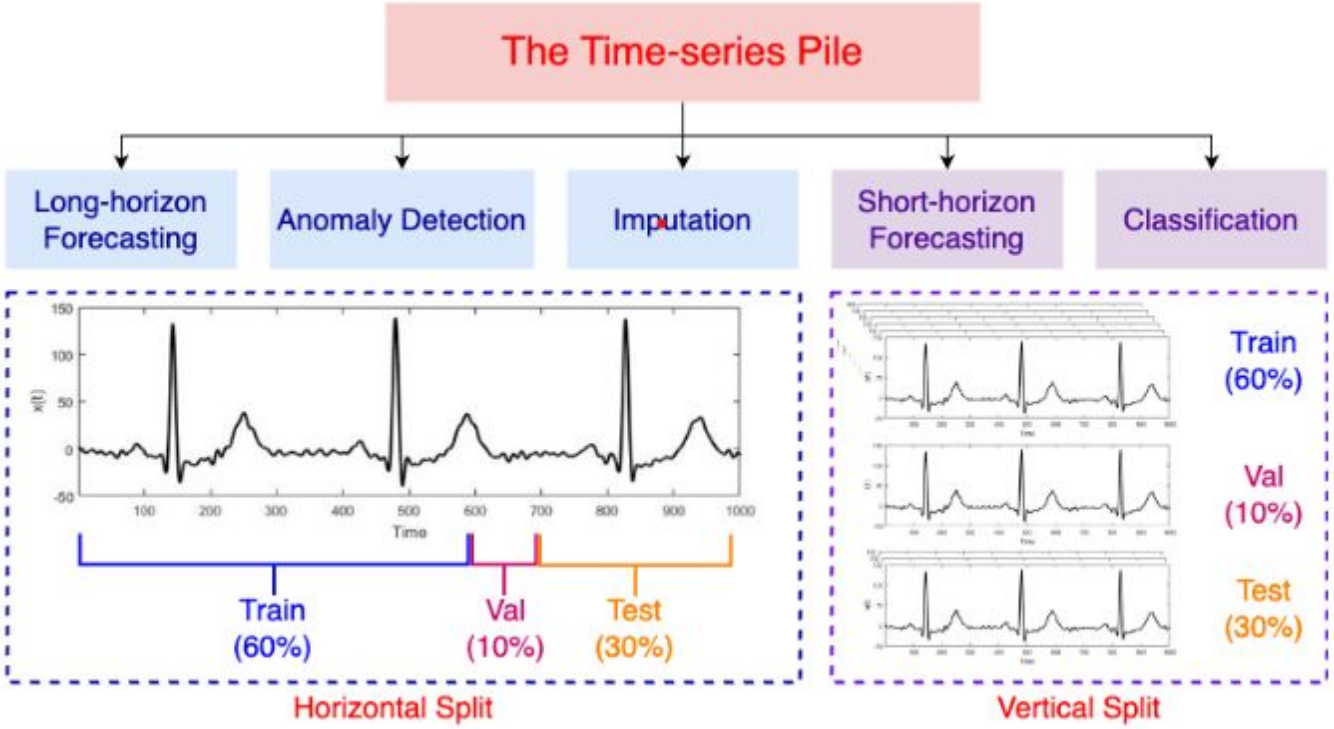
Univariate time series

Length 512, left padded

Absolute + Relative position Embedding

Reconstruction error minimized

Dataset



Collection of 4 public repos that have been widely used

Contributions?
Collation
Data splits provided
Renamed!

Tasks and Experiments

Tasks	Supervision	Datasets	Metrics
Long-horizon Forecasting	Linear Probing	ETT-h1/h2/m1/m2, Electricity, Traffic, Weather, Exchange, ILI	MSE, MAE
Short-horizon Forecasting	Zero-shot	M3 and M4 competition datasets (subset)	sMAPE[4].
Classification	Unsupervised representation learning	UCR Classification Archive	Accuracy
Anomaly Detection	Linear probing, Zero-shot	UCR Anomaly Archive	Adjusted Best F1 VUS-ROC
Imputation	Linear probing, Zero-shot	ETT-h1/h2/m1/m2, Electricity, Weather	MSE, MAE

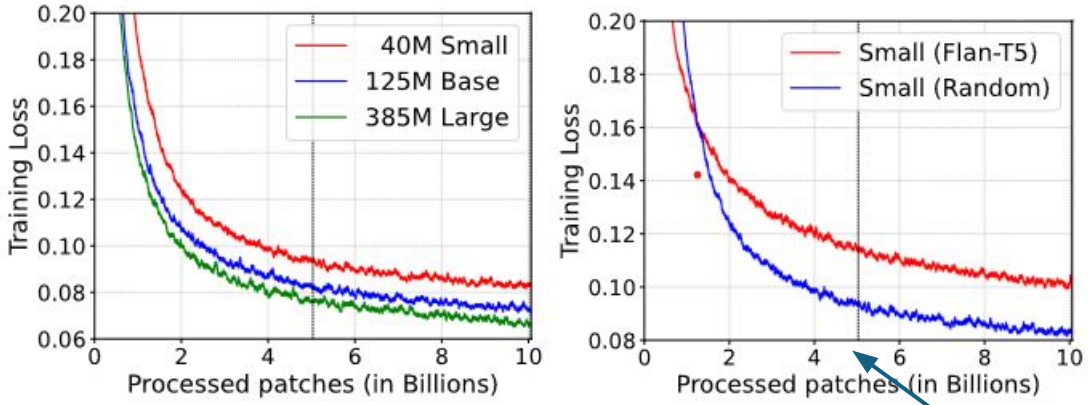
$$\mathrm{SMAPE} = \frac{2}{n}\sum_{t=1}^{n}\frac{|F_t - A_t|}{|A_t| + |F_t|}$$

Results

Long term forecasting: PatchTST is better than MOMENT
Short term forecasting: Mixed (MOMENT, GPT4TS, NBEATS)
Classification: Mixed (? “without finetuning is better”)
Imputation: Mixed
Anomaly detection: Mixed

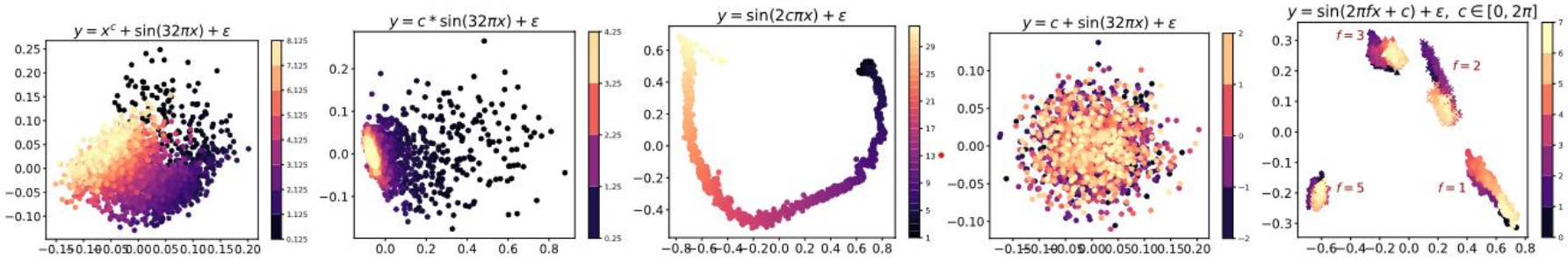
Other methods get domain data specific training.

Further questions



Model scaling improves training loss.

MOMENT with randomly initialized weights converges to a lower training loss. Our observations suggest that with



(i) Trend *(ii)* Amplitude *(iii)* Frequency *(iv)* Baseline Shift *(v)* Phase

What is MOMENT Learning?

MOMENT can solve cross-modal sequence learning tasks. ???